

Sample Project Report for **Biomass Pellets Manufacturing** **Plant**



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1. General Project Information

Particulars	Details
Company Name	Kudra Agro India Private Limited
Contact Person	Vinay Kumar Rastogi
Contact Number	94241 40656
Project Location	Kudra,Bihar
Constitution Of Firm	Proprietorship /LLP/Pvt Ltd.
Nature Of Business	Biomass Pellet Manufacturing
Working Days Per Year	300 Days
Working Hours Per Day	16 Hours
Shift Pattern	2 Shifts (8 Hours Each)

2. Introduction to Biomass Pellets

Particular	Detailed Explanation
Definition	Biomass pellets are solid, cylindrical bio-fuel products manufactured by compressing agricultural residues, forestry waste, and agro-industrial by-products under high pressure. The natural lignin present in biomass acts as a binder, eliminating the need for chemical additives.
Purpose	Designed as a renewable alternative to fossil fuels such as coal, furnace oil, and lignite for industrial thermal applications.
Physical Characteristics	Uniform cylindrical shape; diameter typically 6 mm, 8 mm, or 10 mm; length ranges between 10–30 mm. Smooth surface, high density, and easy handling.
Moisture Content	Maintained between 8%–12% to ensure efficient combustion, reduced smoke emission, and higher calorific value.
Calorific Value	Typically ranges from 3,500 to 4,200 Kcal/kg depending upon the type of raw material used. Higher calorific value ensures better fuel efficiency.
Bulk Density	Approximately 600–750 kg/m ³ , significantly higher than loose biomass, enabling cost-effective storage and transportation.
Ash Content	Normally 5%–10%, depending on agro-residue composition. Lower ash pellets are preferred for boiler efficiency.
Combustion Efficiency	Provides steady and controlled combustion with lower sulfur and nitrogen emissions compared to coal.

Particular	Detailed Explanation
Storage & Handling	Easy to store in bulk form; requires covered storage to prevent moisture absorption. Mechanized feeding systems can be used in industrial boilers.
Applications	Industrial boilers, steam generation plants, textile dyeing units, food processing industries, chemical plants, paper mills, ceramic industries, and co-firing in coal-based thermal power plants.
Environmental Benefits	Carbon-neutral fuel; reduces greenhouse gas emissions; mitigates stubble burning; lowers particulate emissions compared to coal.
Economic Advantages	Lower operational fuel cost compared to furnace oil; price stability compared to fossil fuels; reduces dependency on imported coal.
Policy Support	Promoted under renewable energy initiatives and biomass co-firing programs in thermal power plants.

Strategic Importance

Biomass pellets play a critical role in:

- Transition toward clean and renewable energy sources.
- Reducing carbon footprint of industries.
- Supporting farmers by creating additional revenue streams from crop residues.
- Meeting environmental compliance norms imposed on industrial units.

3. Installed Capacity, Production, Sales & Profit Projection

A. Basic Operating Assumptions

Particular	Details
Working Hours per Day	16 Hours (2 Shifts)
Working Days per Year	300 Days
Total Operating Hours per Year	4,800 Hours
Selling Price	₹7 – ₹14 per Kg
Net Profit Margin	10% – 20%

B. Annual Production Calculation

Formula: Plant Capacity (TPH) × 16 Hours × 300 Days

Plant Capacity	Production per Day (MT)	Annual Production (MT)
1.5 TPH	24 MT	7,200 MT
2.0 TPH	32 MT	9,600 MT
2.5 TPH	40 MT	12,000 MT
4.0 TPH	64 MT	19,200 MT
5.0 TPH	80 MT	24,000 MT

(1 MT = 1,000 Kg)

C. Revenue Projection (Based on Selling Price ₹7–₹14 per Kg)

Plant Capacity	Annual Production (Kg)	Annual Sales @ ₹7/kg	Annual Sales @ ₹14/kg
1.5 TPH	72,00,000 Kg	₹5.04 Cr	₹10.08 Cr
2.0 TPH	96,00,000 Kg	₹6.72 Cr	₹13.44 Cr
2.5 TPH	1,20,00,000 Kg	₹8.40 Cr	₹16.80 Cr
4.0 TPH	1,92,00,000 Kg	₹13.44 Cr	₹26.88 Cr
5.0 TPH	2,40,00,000 Kg	₹16.80 Cr	₹33.60 Cr

D. Estimated Net Profit Projection (10% – 20%)

Plant Capacity	Profit @ 10% Margin	Profit @ 20% Margin
1.5 TPH	₹0.50 Cr	₹2.01 Cr
2.0 TPH	₹0.67 Cr	₹2.68 Cr
2.5 TPH	₹0.84 Cr	₹3.36 Cr
4.0 TPH	₹1.34 Cr	₹5.37 Cr
5.0 TPH	₹1.68 Cr	₹6.72 Cr

E. Cost Structure Overview (Indicative)

Cost Component	Approximate Share (%)
Raw Material	55% – 65%
Electricity & Fuel	8% – 12%
Labor	5% – 8%
Transportation	5% – 10%
Maintenance	3% – 5%
Administrative & Misc.	3% – 5%

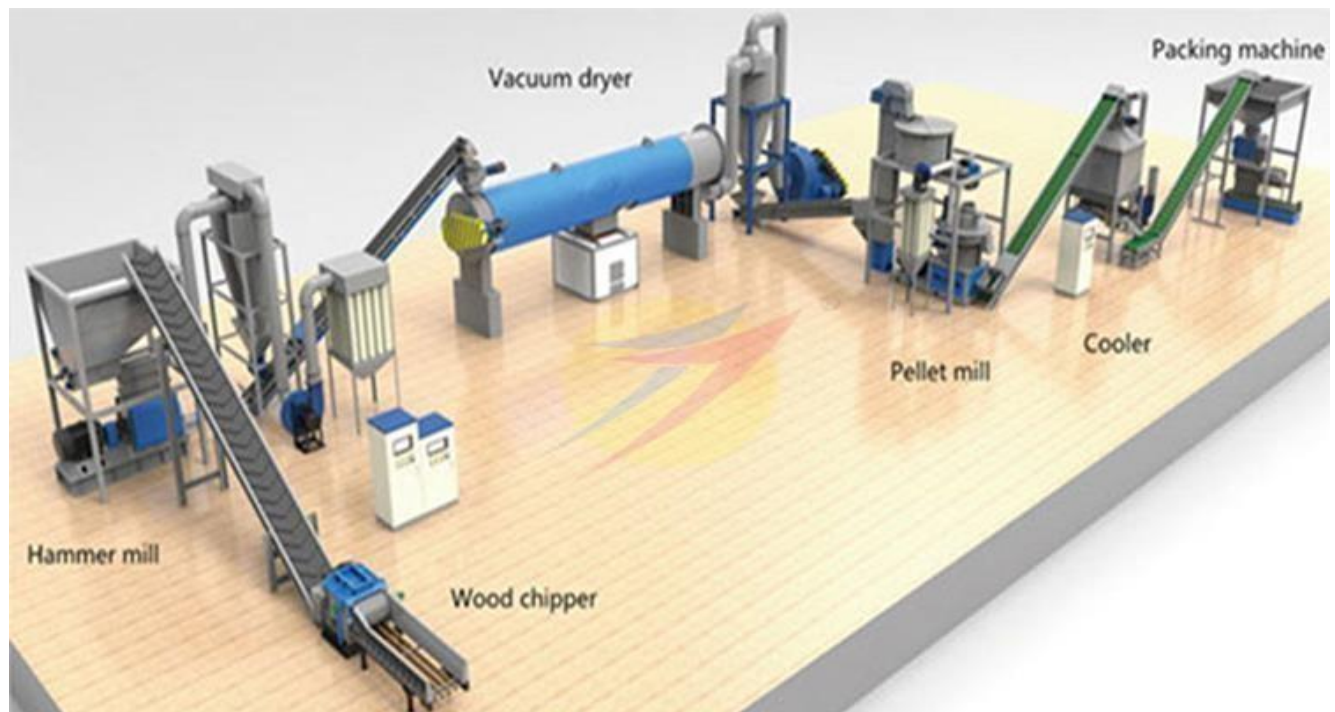
Raw material cost is the most critical profitability factor. Efficient procurement and storage significantly improve margins.

F. Break-Even & Payback Insight

Parameter	Estimate
Break-Even Point	50% – 60% Capacity Utilization
Payback Period	3 – 5 Years

G. Financial Strength Indicators

- Higher capacity plants achieve better economies of scale.
- Long-term supply contracts reduce price volatility risk.
- Stable demand from industrial boiler segment ensures recurring revenue.
- Profit margins improve with optimized logistics and direct farmer sourcing.



4. Manufacturing Process

The manufacturing process of biomass pellets involves systematic conversion of loose agricultural and forestry residues into high-density solid fuel through mechanical processing. The process ensures uniform size, low moisture content, high calorific value, and efficient combustion properties.

A. Detailed Process Description

Stage	Process Description
1. Raw Material Procurement & Storage	Agricultural residues such as paddy straw, sawdust, husk, cotton stalk, etc., are collected from farmers, sawmills, or agro-processing units and stored in covered sheds.
2. Pre-Cleaning (if required)	Removal of stones, metal particles, and impurities using magnetic separators and screens.
3. Drying	Raw biomass usually contains 15–30% moisture. A rotary drum dryer reduces moisture to 8–12%. Heat source may be biomass furnace or hot air generator.
4. Grinding / Size Reduction	Dried material is fed into a hammer mill to reduce particle size to 3–5 mm.
5. Pelletizing	Ground material is fed into a pellet mill. Under high pressure and temperature, rollers compress the material through a die to form cylindrical pellets. Natural lignin binds particles.
6. Cooling	Fresh pellets exit at high temperature (70–90°C). A counter-flow cooler reduces temperature to near ambient level.
7. Screening	Vibrating screens separate fines and broken pellets, which are recycled back to the pellet mill.
8. Packing / Bulk Handling	Finished pellets are packed in 25 kg / 50 kg bags or loaded in bulk trucks using conveyors.

B. Machinery Involved

Equipment	Function
Rotary Dryer	Moisture reduction
Hammer Mill	Size reduction
Pellet Mill	Pellet formation
Counter Flow Cooler	Temperature control
Vibrating Screen	Removal of fines
Belt Conveyor / Screw Conveyor	Material transfer
Dust Collector System	Pollution control
Control Panel (PLC Based)	Automation and monitoring

C. Technical Parameters

Parameter	Standard Range
Pellet Diameter	6–10 mm
Pellet Length	10–30 mm
Moisture Level	8–12%
Bulk Density	600–750 kg/m ³
Calorific Value	3,500–4,200 Kcal/kg

D. Power & Utility Requirement

Particular	Requirement (Approx.)
Connected Load	200–600 HP (depending on plant capacity)
Fuel for Dryer	Biomass waste
Water Requirement	Minimal (domestic use only)

5. Project Cost & Means of Finance

Particulars	Amount (₹ in Lakhs)
Construction	160
Pellet Manufacturing Plant	250.00
Internal and External electrification	15.00
MFA	5.00
DP/Transformer	85.00
Weigh Bridge	10.00
Furniture	5.00
Solar System	NA
Loader	25.00
Total Fixed Assets	555.00
Working Capital	150.00
Total Project Cost	705.00

Means of Finance

Source	Amount (₹)
Own Contribution (25%)	138.75
Term Loan (75%)	416.25
Working Capital (100%)	150.00
Total	705.00

6. Raw Material Availability

Raw material is the most critical component in biomass pellet manufacturing, contributing approximately **55%–65% of total production cost**. The availability, price stability, moisture content, and logistics of biomass directly impact profitability and plant efficiency.

India, being an agriculture-dominant country, generates large quantities of crop residues and agro-industrial waste, ensuring abundant raw material availability throughout the year.

A. Major Categories of Raw Materials

1. Agricultural Residues

Raw Material	Source	Availability Season	Remarks
Paddy Straw	Rice-growing regions	Oct–Jan	Highly abundant; widely used
Wheat Straw	Agricultural fields	Apr–Jun	Seasonal but bulk available
Cotton Stalk	Cotton-growing areas	Dec–Mar	Good calorific value
Maize Cobs	Corn processing areas	Seasonal	Uniform structure
Mustard Husk	Oil mills	Seasonal	Moderate ash content
Soy Husk	Soybean processing units	Year-round	Consistent supply
Groundnut Shell	Oil extraction units	Seasonal	Good combustion
Sunflower Husk	Oil mills	Seasonal	Medium ash content
Sugarcane Trash	Sugar factories	Crushing season	Requires drying
Bagasse	Sugar mills	Seasonal	Moisture control required

2. Forestry & Wood Residues

Raw Material	Source	Availability
Sawdust	Sawmills	Year-round
Wood Chips	Timber industries	Regular
Bamboo Dust	Bamboo processing units	Regular
Wood Shavings	Furniture industries	Regular

Wood-based materials generally have lower ash content and higher calorific value compared to agro residues.

3. Agro-Industrial Waste

Raw Material	Source
Coffee Husk	Coffee processing units
Tea Waste	Tea processing industries
Rice Husk	Rice mills
Pulse Husk	Dal mills
Coconut Shell Powder	Coconut processing units

These materials are often available in concentrated industrial clusters, reducing transportation cost.

B. Raw Material Characteristics

Parameter	Ideal Range
Moisture Content (Before Drying)	15%–30%
Moisture After Drying	8%–12%
Particle Size (After Grinding)	3–5 mm
Ash Content	5%–10%
Calorific Value	3,000–4,200 Kcal/kg

Lower moisture and ash content improve pellet durability and combustion efficiency.

C. Geographic Availability in India

- **North India:** Paddy straw, wheat straw
- **Central India:** Soy husk, cotton stalk
- **Western India:** Groundnut shell, cotton waste
- **Southern India:** Bagasse, coffee husk, rice husk

The wide distribution of agro residues ensures flexibility in plant location.

D. Procurement Strategy

Strategy	Benefit
Direct Purchase from Farmers	Lower cost, better margins
Tie-up with FPOs / Cooperatives	Bulk and steady supply
Contracts with Oil Mills / Rice Mills	Continuous availability
Seasonal Bulk Storage	Protection against price fluctuations

Long-term agreements with suppliers reduce price volatility risk.



7. Market Research

Market research for biomass pellets focuses on demand analysis, customer segments, pricing trends, competition assessment, regulatory environment, and future growth prospects. The biomass pellet industry is primarily a **B2B (Business-to-Business)** market driven by industrial fuel requirements and environmental compliance norms.

A. Target Customer Segments

Industry Segment	Application	Fuel Replacement
Thermal Power Plants	Co-firing with coal	Partial coal replacement
Textile Industries	Steam boilers	Coal / Furnace Oil
Food Processing Units	Steam generation	Coal / FO
Paper & Pulp Mills	Boiler fuel	Coal
Chemical Industries	Process heating	FO / Lignite
Ceramic & Tile Units	Kiln heating	Coal
Brick Kilns	Firing fuel	Coal
Dairy & Agro Industries	Steam boilers	Furnace Oil

Industrial boilers represent the largest and most stable demand segment.

B. Demand Drivers

1. **Rising Coal Prices** – Increasing cost of imported and domestic coal makes biomass pellets competitive.
 2. **Environmental Regulations** – Industries must reduce carbon emissions and particulate matter.
 3. **Government Co-firing Mandates** – Coal-based power plants are encouraged to blend biomass.
 4. **Corporate ESG Goals** – Companies adopting sustainable energy practices.
 5. **Pollution Control Norms** – Restrictions on high-sulfur fuels like furnace oil.
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C. Market Growth Trend

Parameter	Estimate
Industry Growth Rate	15%–20% annually
Demand Trend	Strong upward trajectory
Industrial Adoption	Increasing year-on-year
Renewable Energy Push	Expanding rapidly

The renewable energy transition is significantly strengthening biomass pellet demand across India.

D. Pricing Analysis

Factor	Impact on Price
Raw Material Cost	Major determinant
Transportation	Influences delivered price
Calorific Value	Higher CV fetches better rate
Ash Content	Lower ash preferred
Seasonal Demand	Winter demand slightly higher

Typical selling price range: ₹7 – ₹14 per Kg depending on region, quality, and bulk contracts.

E. Supply Chain Structure

Manufacturer → Transporter → Industrial Buyer

Or

Manufacturer → Direct Long-Term Contract → Power Plant / Industry

Direct contracts improve profit margins and reduce intermediary costs.

F. Buyer Purchase Criteria

- Consistent calorific value
 - Low moisture content
 - Timely delivery
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- Competitive pricing
- Reliable long-term supply

Industries prefer suppliers who can maintain steady quality and uninterrupted supply.

G. Long-Term Market Outlook

- Stable and growing industrial demand
- Government-backed renewable initiatives
- Increasing fossil fuel substitution
- Strong potential for multi-year supply agreements

8. Business & Market Potential

The biomass pellet manufacturing business offers strong long-term potential due to increasing industrial fuel demand, renewable energy policies, and rising fossil fuel costs. The sector combines environmental sustainability with stable commercial returns, making it an attractive investment opportunity.

A. Industry Outlook

Parameter	Details
Industry Nature	Renewable Energy / Industrial Fuel
Growth Rate	15% – 20% annually (estimated)
Demand Stability	High and recurring (boiler-based consumption)
Policy Support	Strong government backing for biomass co-firing
Entry Barrier	Moderate (capital investment required)

The industry is categorized as a sunrise sector under the renewable energy ecosystem.

B. Revenue Potential

Revenue depends on plant capacity, operational efficiency, and selling price.

Plant Capacity	Approx. Annual Revenue Range
1.5 TPH	₹5 – ₹10 Crore
2.0 TPH	₹6 – ₹13 Crore
2.5 TPH	₹8 – ₹16 Crore
4.0 TPH	₹13 – ₹26 Crore
5.0 TPH	₹16 – ₹33 Crore

Higher capacities generate better economies of scale and improved per-unit profitability.

C. Profitability Potential

Parameter	Estimate
Net Profit Margin	10% – 20%
Break-Even Point	50% – 60% Capacity Utilization
Payback Period	3 – 5 Years
Return on Investment (ROI)	Attractive (subject to cost efficiency)
Operating Margin Stability	Moderate to High

Efficient raw material sourcing and optimized logistics significantly improve margins.

D. Demand Sustainability

1. **Industrial Boiler Dependence** – Industries require continuous steam generation, ensuring regular pellet consumption.
2. **Thermal Power Plant Co-firing** – Biomass blending with coal ensures bulk demand.
3. **Environmental Compliance** – Industries switching to cleaner fuels to meet pollution norms.
4. **Long-Term Supply Contracts** – Ensures predictable cash flow.

Demand is not seasonal and remains steady throughout the year.

E. Scalability & Expansion Opportunity

Expansion Option	Benefit
Adding Additional Pellet Line	Increase capacity with moderate incremental investment
Forward Integration	Supplying directly to large power plants
Backward Integration	Raw material aggregation from farmers
Export Market Entry	Higher margins for premium-grade pellets

The modular structure of pellet plants allows phased expansion.

9. Subsidy & Government Support

A. Agriculture Infrastructure Fund (AIF)

Objective: Financing for post-harvest management and agri-infrastructure projects including biomass aggregation and processing units.

Particular	Details
Interest Subvention	3% per annum on term loan
Loan Limit Eligible for Subvention	Up to ₹2 Crore
Duration of Benefit	Maximum 7 Years
Credit Guarantee	Coverage available under CGTMSE (as applicable)
Moratorium	As per bank norms
Eligible Entities	Individuals, FPOs, Agri-entrepreneurs, Companies, Cooperatives

Impact on Project:

- Reduces effective interest rate by 3%.
- Lowers repayment burden in initial years.
- Improves project profitability and cash flow.

B. Ministry of New & Renewable Energy (MNRE)

Objective: Promotion of biomass energy and clean fuel adoption.

Particular	Details
Central Financial Assistance (CFA)	Capital subsidy (percentage varies as per scheme notification)
Focus Area	Biomass briquette and pellet manufacturing units
Applicable Projects	Units supplying biomass for co-firing or renewable use
Mode of Assistance	Through implementing agencies or state nodal agencies

C. State Renewable Energy Subsidy

Most states provide additional capital subsidy under their Renewable Energy or MSME policies.

Particular	Typical Range
Capital Subsidy	Eligible for capital fix assets or plant & machinery cost (applicable as per state policy)
Maximum Limit	As per state policy
Electricity Duty Exemption	Available in some states
Stamp Duty Reimbursement	Available in selected states
SGST Reimbursement	As per industrial policy

E. Other Financial Benefits

Benefit	Description
Income Tax Benefits	As per applicable tax laws
Accelerated Depreciation	As per Income Tax provisions
Priority Sector Lending	Eligible under renewable energy category
Carbon Credit Potential	Subject to eligibility and certification

F. Combined Subsidy Impact Example

Type of Benefit	Financial Advantage
3% Interest Subvention (AIF)	Reduces EMI burden
20% Capital Subsidy (State Example)	Reduces initial capital cost
CGTMSE Coverage	Reduces collateral requirement

These incentives significantly improve project bankability and shorten the payback period.

11. Conclusion

A. Technical Feasibility

- Proven and mature pelletizing technology.
- Fully automated and continuous production process.
- Easy scalability by adding additional pellet lines.
- Availability of skilled technical manpower.
- Standardized machinery with low maintenance requirements.

The manufacturing process is simple, systematic, and adaptable to various types of biomass raw materials.

B. Raw Material Strength

- Abundant agricultural residue availability across India.
- Diverse raw material base reduces dependency on a single source.
- Opportunity to establish direct farmer linkages.
- Supports waste-to-energy model and circular economy.

Strong raw material availability ensures long-term operational sustainability.

C. Financial Viability

Parameter	Assessment
Revenue Potential	High (₹5 Cr – ₹33 Cr depending on capacity)
Net Profit Margin	10% – 20%
Break-Even	50% – 60% capacity utilization
Payback Period	3 – 5 years
ROI Potential	Attractive

The project demonstrates strong earning capacity with scalable revenue and stable operating margins.

D. Market Sustainability

- Growing demand from industrial boiler segment.
- Increasing adoption by thermal power plants for co-firing.
- Rising fossil fuel prices improving competitiveness.
- Supportive renewable energy policies.

The demand outlook is stable and expected to grow steadily over the next decade.

E. Government & Policy Support

- Interest subvention benefits under AIF.
- Capital subsidy opportunities under renewable energy schemes.
- MSME benefits including credit guarantee and financial support.

Policy backing enhances project bankability and reduces financial risk.

F. Environmental & Social Impact

Impact Area	Benefit
Pollution Control	Reduction in stubble burning
Carbon Emissions	Lower greenhouse gas emissions
Rural Employment	Direct & indirect job creation
Farmer Income	Additional revenue stream
Sustainable Energy	Contribution to renewable energy targets

The project aligns with national clean energy and sustainability goals.

G. Long-Term Outlook

- Increasing industrial shift toward clean fuel.
- Growing ESG compliance requirements.
- Potential for export-grade pellet production.
- Strong opportunity for capacity expansion and integration.

The biomass pellet sector is expected to remain a high-growth renewable energy segment.
